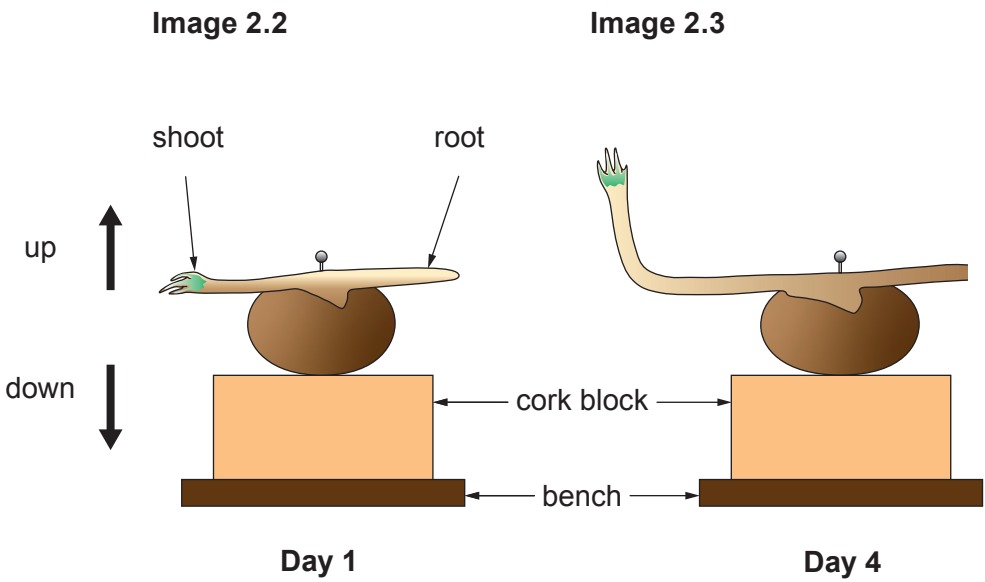


(b) **Images 2.2 and 2.3** show an investigation into the effect of gravity on the growth of a seedling which has been placed horizontally on a laboratory bench.

(i) **Complete Image 2.3** by drawing **the root** as you would expect to see it at **day 4**. [2]



(ii) I. State the name of the response of the root to gravity. [1]

II. Explain why this response is important to the seedling. [1]

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4. In the 1980s young terrapins (*Trachemys scripta*) from North America were sold in pet shops throughout the UK. When the terrapins became adults, many people no longer wanted them and released them into local ponds. The animals can live for up to 50 years.



Young terrapin



Adult terrapin

Table 4.1

Terrapin age	Body length (mm)
3 months	25
2 years	50
12 years	300

After release, adult terrapins eat native animals including invertebrates and vertebrates such as frogs and fish. In some ponds, native species have been reduced or lost completely. Since 1992 a law has banned the import of terrapins into the UK.

Use the information to answer the questions.

- (a) Complete the classification of the terrapin.

[3]

Kingdom
Phylum Chordates (Vertebrates)
Class Reptiles
Genus
Species



- (b) (i) Calculate the increase in body length **per year** for an adult terrapin in the **ten years** between 2 years old and 12 years old. [2]

Increase in body length = mm/year

- (ii) Suggest **one** reason why some people did not want to keep the terrapins when they became too big. [1]

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- (c) **Complete Table 4.2** by writing true or false for each of the statements. One row has been done for you. [3]

Table 4.2

Statement about terrapins in the UK	True / False
They can live for over 60 years.	False
They do not eat animals with backbones.	
They have reduced the biodiversity in ponds.	
They cannot be brought into the country legally.	
They are an alien species.	

- (d) In the USA, these terrapins are called red-eared sliders and in Welsh they are called terapin clustgoch.

State how scientists from different countries ensure that they are all talking about the same species. [1]

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